Nicholas Pianetto

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EDUCATION

Iowa State University, College of Engineering **Expected December 2026**
Bachelor of Science, Computer Engineering

SKILLS

Programming Languages: C/C++, Python, Verilog, VHDL, 32-bit MIPS Assembly, Java, HTML, CSS

Embedded/Hardware: Arduino, STM32, ESP32, Xilinx Zynq-7000, CAN/J1939, SPI, I²C, UART, Soldering/Rework, Multimeter, Logic Analyzer, Oscilloscope

EDA/Software: PCB Design (PADS Logic/Layout, KiCad, Altium Designer, Zuken), STM32CubeIDE, Lattice Diamond, SolidWorks, Linux, PetaLinux, Vivado, Vitis, QuestaSim, Quartus Prime, LTspice, Git, Perforce

WORK EXPERIENCE

Electrical Engineer Intern - *Vermeer Corporation* **May 2026 - August 2026**
- Designed a PCB for a wireless ESP32-based CAN bus device to modify engine RPM during off-machine system testing, eliminating the need for a $350 tethered CAN-to-USB interface
- Reduced setup and testing time of a prototype machine harness by nearly 50% by integrating CAN bus controllers and a Linux-based display module for verification of machine harness pinouts
- Designed and shipped a cable harness adapter to maintain pin compatibility for vendor part revision changes
- Conducted field testing on a machine to measure and log inrush current for cooling systems and mapped data against interconnect thermal limits to develop a mitigation strategy for overcurrent failures
- Designed the schematic in Altium Designer for a Renesas RA4M1-based system and generated a BOM for cost estimation

Circuits II Teaching Assistant - *Iowa State University* **January 2026 - May 2026**
- Mentored students in lab with hands-on debugging, instrumentation guidance, and circuit synthesis
- Supported grading homework assignments and quizzes involving analog circuitry

Research Assistant - *Iowa State University* **May 2025 - May 2026**
- Independently designed the schematic and PCB layout, and wrote firmware for a bioimpedance device that cost less than $30, successfully replacing an $8,000 device, utilizing the on-chip DFT of the AD5933 high-precision impedance converter
- Conducted meetings with stakeholders from the Department of Animal Science to establish design requirements such as built-in SPI-based SD card data logging and a portable LCD interface and wrote the firmware required to drive both features

Embedded Design Engineer Intern - *Ag Leader Technology* **May 2024 - December 2024**
- Independently developed the schematic and PCB layout for a passive USB Type-C PCB with impedance matching on all differential pairs to support DisplayPort functionality for product testing during assembly
- Revised a serial to fiber optic PCB in PADS Layout and PADS Logic for use during ESD, EMC, and EMI testing
- Reworked prototype PCBs by soldering experimental shielding and 0402-sized bypass capacitors in an effort to minimize capacitive coupling problems during ESD testing and successfully mitigated identified issues

PROJECTS

PONG on Custom FPGA Development Board with VGA
- Independently designed the schematic, layout, assembled, and programmed a custom PCB based on the Lattice MachXO2 7000HC
- Integrated onboard power regulation, configurable I/O voltage banks, synchronous 640x480 VGA timing logic, a custom FTDI FT2232H-based JTAG programmer, and expansion headers; implemented PONG in VHDL (see https://pianetto.net/projects)

CPRE 488 Open-Ended Design Project (MIDI Music On Hardware)
- Developed a custom memory-mapped state-machine IP block for PWM and 4-phase motor control in VHDL, enabling MIDI playback through floppy drive motors using a Xilinx Zedboard running Linux
- Reverse-engineered a flatbed scanner PCB by analyzing datasheets of ICs and modifying existing circuitry to directly control the 4-phase motor from the Zedboard's PMODs using onboard Darlington transistor arrays and external logic level shifters
- Rewrote and optimized C++ drivers to remove dependencies on large libraries, successfully shrinking the software footprint from 550 KB to under 250 KB to fit the FSBL with an I²C/SPI-based MIPI camera within the limited OCM of the Zedboard